

it is critical that the power consumption is minimised to increase battery life.

Where to look?

Medical Electronics & Telemedicine Division, Department of Information Technology (DIT), Ministry of Communications & Information Technology. Under the aegis of DIT, a 6MeV medical LINAC (linear accelerator) used for radiation therapy in cancer patients has been installed at Mahatma Gandhi Institute of Medical Sciences, Wardha, and RCC, Adyar, Chennai. Besides, a lot of technology products have been developed for the visually challenged people and successfully implemented through a number of blind schools in the country.

DIT has also undertaken development of telemedicine technology to provide quality healthcare to patients in rural/remote locations through specialists. Under telemedicine pilot projects, telemedicine centres have been set up in Tripura, Punjab, Himachal Pradesh, West Bengal, Kerala and Tamil Nadu. Several doctors and paramedics have also been trained on using this technology.

Larsen & Toubro (L&T). Larsen & Toubro (L&T) had commenced its medical business operations in 1988, with the core aim of bringing contemporary healthcare equipment technology to India and making medical equipment cost-effective over their life-cycle.

Its state-of-the-art factory at Mysore manufactures patient monitoring systems, ultrasound scanners and electrosurgical units for India and the global market. More than 25 models have been launched during the last decade by its in-house design and development team. Most of these models have received the coveted CE mark and many of its patient monitors have been cleared for exports to US by the US FDA. L&T Medical also has traded lines specifically for anaesthesia



Larsen & Toubro medical devices

products, syringe infusion pumps and biphasic defibrillators.

L&T Medical has domain expertise in critical care devices and more specifically in patient monitoring systems that are used primarily in intensive-care units and operation theatres to measure vital parameters of patients.

BPL. BPL claims to be the market leader in the single-channel ECG market with 72 per cent share. In 2009-10, it strengthened its market share in the ECG market with significant contribution from three-channel and six-channel categories. It expanded its Cardiart ECG range with new models in three-channel and six-channel segments to garner more than 55 per cent share in three-channel ECG segment and more than 65 per cent share in six-channel ECG segment.

Recently, the company introduced a slew of products for six-channel, 12-channel and Holter with features focused on clinical productivity and network connectivity. The new BPL Holter, TRAK48, is very compact

and has patient-friendly features and extensive interpretation software for clinicians.

MindTree. MindTree is working on devices and solutions to overcome the affordability and accessibility challenges for Indian healthcare delivery. Portable ultrasound scanner is being developed along with various applications for

cardiac disease detection. The cardiac disease detection tool can also work offline, so doctors can collaborate and consult amongst themselves to serve the patients better.

MindTree's MindTMED telemedicine platform combines the flexibility of remote conferencing and VoIP to deliver a simple solution-in-a-box that optimises any healthcare service delivery across geographies. It brings the patient and doctor closer, at any time of the day, irrespective of their time zones. This is achieved by integrating videoconferencing and communication technologies into a single box. The platform takes inputs from common medical devices (like ECG, NiBP, temperature and SPO2) through its connectivity options and transmits them to a remote doctor's site. This allows the doctor to view the data of the patient on a PC or laptop, and then make a diagnosis if the ailment is common or if it requires special treatment.

With extensive short-range wireless experience, MindTree also provides ef-

“Majority of medical electronic devices broadly contain chips that capture the input data and convert it for processing, digital processing chips that process the data, and imaging chips for display of diagnostic results and video during complex surgical procedures—Alok Mehrotra, managing director, Magma Design Automation India

